

Trees

Simple Tree

Empty tree:

Tree with one node:

Fred

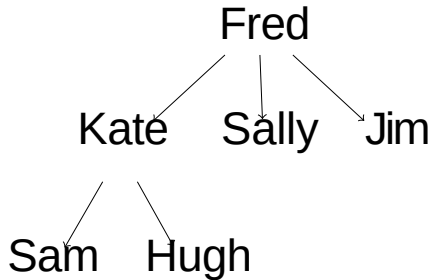
Tree with two nodes:

Fred

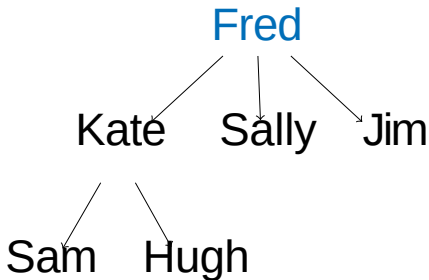


Sally

Terminology

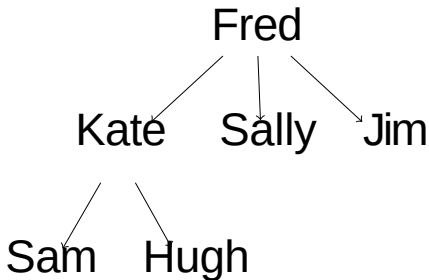


Terminology



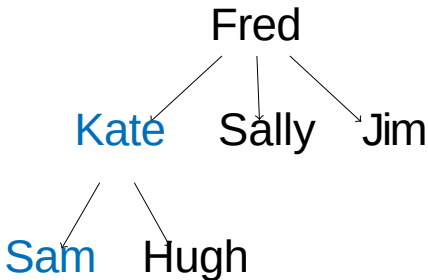
Root:
top node in the tree

Terminology



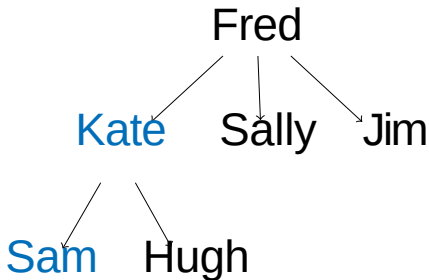
A child has a line down directly from a parent

Terminology



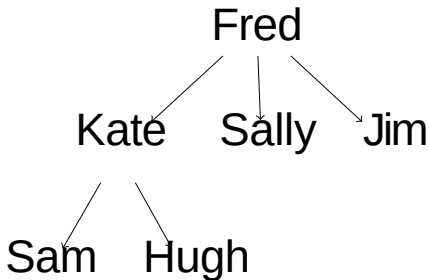
Kate is a *parent* of Sam

Terminology



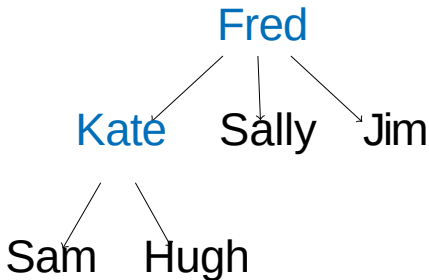
Sam is a *child* of Kate

Terminology



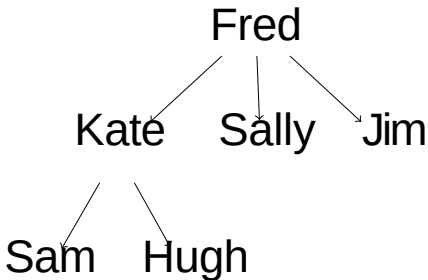
Ancestor:
parent, or parent of parent, etc.

Terminology



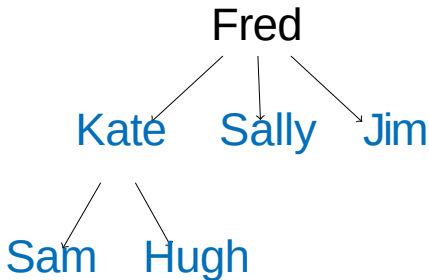
Ancestors of Sam

Terminology



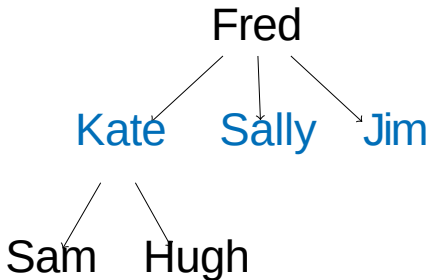
Descendant:
child, or child of child, etc.

Terminology



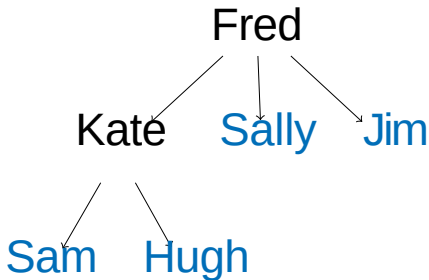
Descendants of Fred

Terminology



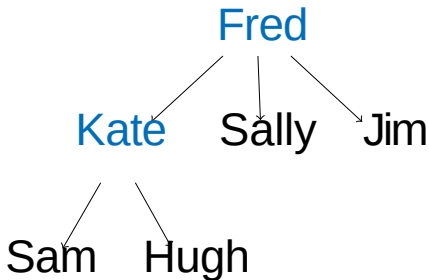
Sibling :
sharing the same parent

Terminology



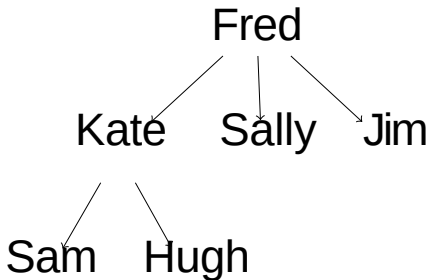
Leaf :
node with no children

Terminology



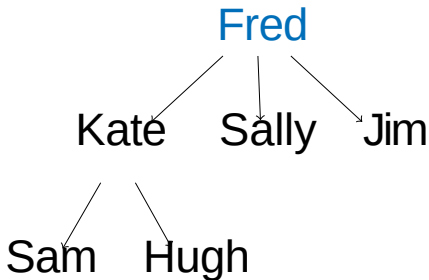
**Interior node
(non-leaf)**

Terminology



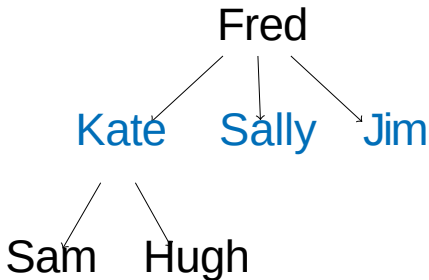
Level: num edges between root
and node

Terminology



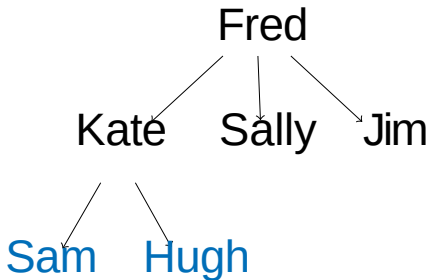
Level 0

Terminology



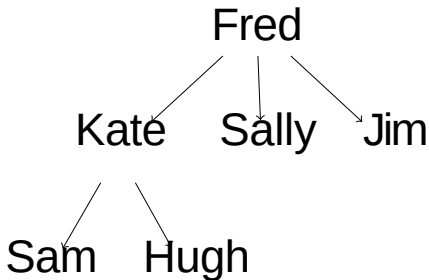
Level 1

Terminology



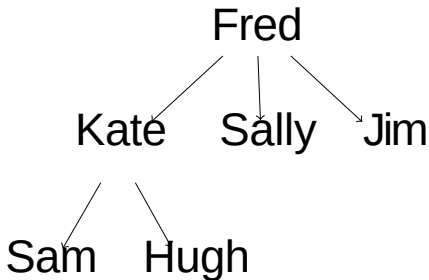
Level 2

Terminology



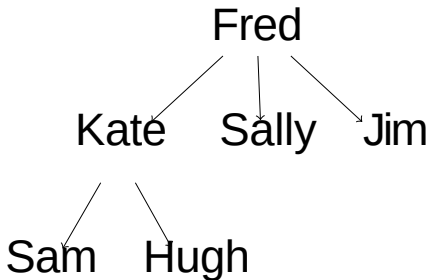
The height of a node is the number of edges from the node to the deepest leaf.

Terminology



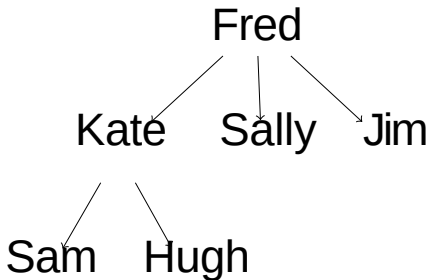
The height of a node is the number of edges from the node to the deepest leaf.

Terminology



The height of a tree is a height of the root.

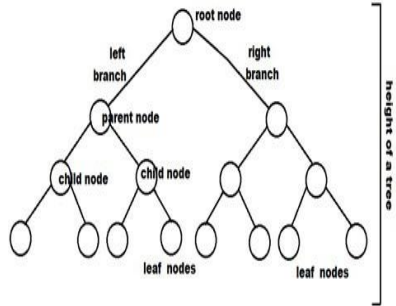
Terminology



The depth of a node is the number of edges from the root to the node.

Binary Tree

- ❑ A binary tree is a data structure composed of nodes, where each node contains a **value** and **two references to other nodes**, called the left child and the right child.
- ❑ In a binary tree each node can have n children (**$0 \leq n \leq 2$**)

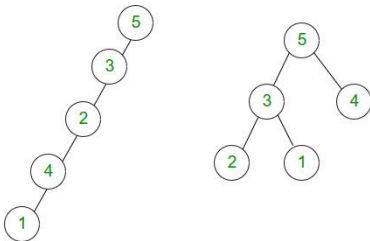


Number of nodes and height of binary tree

Calculating minimum and maximum height from a number of nodes –

If there are n nodes in a binary tree, the maximum height of the binary tree is $n-1$, and the minimum height is $\text{floor}(\log_2(n))$.

For example, the left skewed binary tree shown in Figure below(left) with 5 nodes has a height of $5-1 = 4$, and the binary tree shown in Figure below(right) with 5 nodes has a height $\text{floor}(\log_2 5) = 2$.



Number of nodes and height of binary tree

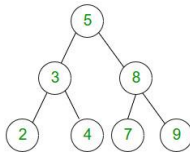
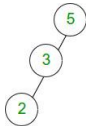
Calculating minimum and maximum number of nodes from height:

If binary tree has height h , minimum number of nodes is $h+1$ (in case of left skewed and right skewed binary tree).

For example, the binary tree shown in Figure 2(a) with height 2 has 3 nodes.

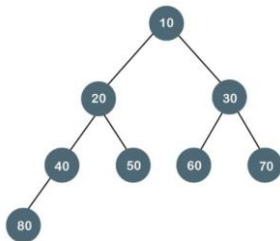
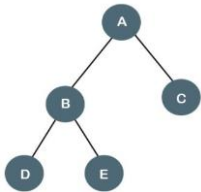
If binary tree has height h , maximum number of nodes will be when all levels are completely full. Total number of nodes will be $2^0 + 2^1 + \dots + 2^h = 2^{(h+1)} - 1$.

For example, the binary tree shown in Figure 2(b) with height 2 has $2^{(2+1)} - 1 = 7$ nodes.



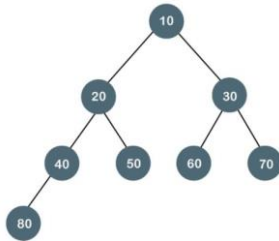
Types of Binary tree

- ❑ Full binary tree: A full binary tree is a binary tree in which every node has **either two children or no children**.
 - ❑ each node must contain 2 children except the leaf nodes.



- ❑ Complete binary tree: A complete binary tree is a binary tree in which all levels except the last are completely filled
 - ❑ All of the nodes must be as far to the **left as feasible** in the **last level**. The nodes should be added from the left

Complete Binary tree



- ❑ Complete binary tree: A complete binary tree is a binary tree in which all levels except the last are completely filled
 - ❑ All of the nodes must be as far to the **left as feasible** in the **last level**. The nodes should be added from the left

A complete binary tree is very special tree, it provides the best possible ratio between the number of nodes and the height. The height h of a complete binary tree with N nodes is at most $O(\log N)$. We can easily prove this by counting nodes on each level, starting with the root, assuming that each level has the maximum number of nodes:

$$n = 1 + 2 + 4 + \dots + 2^{h-1} + 2^h = 2^{h+1} - 1$$

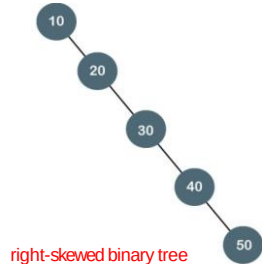
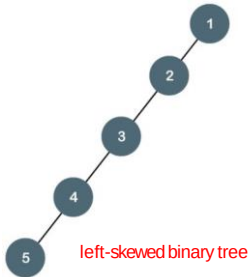
Solving this with respect to h , we obtain

$$h = O(\log n)$$

Types of Binary tree

Skewed binary tree

- A **left-skewed binary tree** is a tree in which each node has at most one right child.
- A right-skewed binary tree is a tree in which each node has at most one left child.



Types of Binary tree

Balanced binary tree: A balanced binary tree is a binary tree in which the heights of the left and right subtrees of every node differ by at most one.

Some operations in a an unbalanced tree may take longer than $O(\log n)$ time

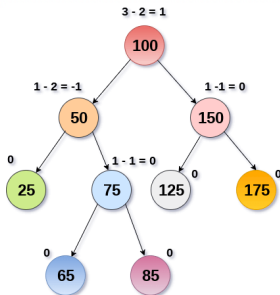
**AVL Tree is a well-known data structure to generate/maintain
Balanced Binary Search Tree.**

Types of Binary tree

AVL Tree can be defined as height balanced binary search tree in which each node is associated with a balance factor which is calculated by subtracting the height of its right sub-tree from that of its left sub-tree.

Tree is said to be balanced if balance factor of each node is in between -1 to 1, otherwise, the tree will be unbalanced and need to be balanced.

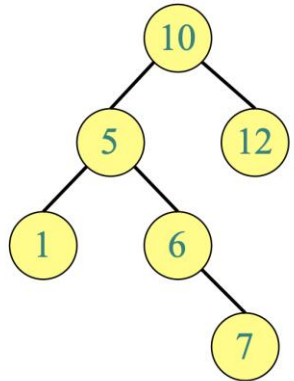
Balance Factor (k) = height (left(k)) - height (right(k))



AVL Tree

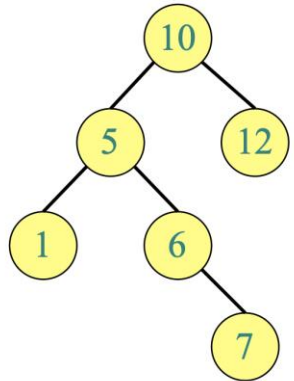
Binary Tree

- Each node x has:
 - $\text{key}[x]$
 - Pointers:
 - $\text{left}[x]$
 - $\text{right}[x]$
 - $\text{p}[x]$



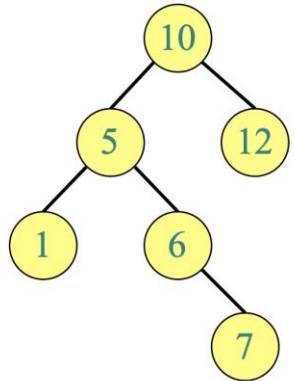
Binary Search Tree

- Each node x has:
 - $\text{key}[x]$
 - Pointers:
 - $\text{left}[x]$
 - $\text{right}[x]$
 - $\text{p}[x]$



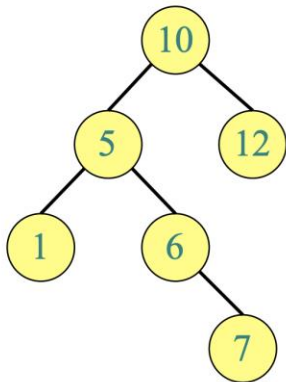
Binary Search Tree

- Each node x has:
 - $\text{key}[x]$
 - Pointers:
 - $\text{left}[x]$
 - $\text{right}[x]$
 - $\text{p}[x]$



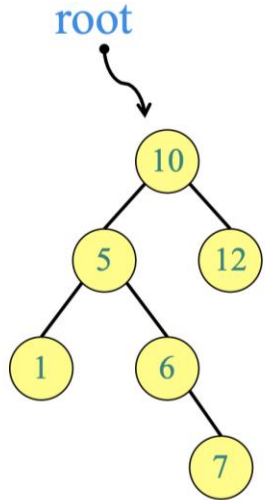
Binary Search Tree

- Property: for any node x :
 - For all nodes y in the **left** subtree of x :
$$\text{key}[y] \leq \text{key}[x]$$
 - For all nodes y in the **right** subtree of x :
$$\text{key}[y] \geq \text{key}[x]$$
- How are BSTs made ?



Growing BSTs

- Insert 10
- Insert 12
- Insert 5
- Insert 1
- Insert 6
- Insert 7



Number of nodes and height of BST

All the rules in BST are same as in binary tree and can be visualized in the same way.

Calculating minimum and maximum height from the number of nodes:

If there are n nodes in a binary search tree, maximum height of the binary search tree is $n-1$ and minimum height is $\text{ceil}(\log_2(n)+1)$.

Calculating minimum and maximum number of nodes from height:

If binary search tree has height h , minimum number of nodes is $h+1$ (in case of left skewed and right skewed binary search tree).
If binary search tree has height h , maximum number of nodes will be when all levels are completely full. Total number of nodes will be $2^0 + 2^1 + \dots + 2^h = 2^{(h+1)} - 1$.

Question 1

Question 1: Suppose you have a binary search tree (BST) with 31 nodes.

What is the maximum possible height of the tree?

What is the minimum possible height of the tree?

Question 2: You are given a binary search tree with a height of 5.

What is the minimum number of nodes the tree can have?

What is the maximum number of nodes the tree can have?

Question 3: If a binary search tree has 63 nodes, what is the height of the tree when it is perfectly balanced?

Question 2

For the set of $\{1, 4, 5, 10, 16, 17, 21\}$ of keys, draw binary search trees of heights 2, 3, 4, 5, and 6.